The Friedmann Geometric Factor: Multiple Cosmological Relationships

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Abstract

We report the discovery that five independent cosmological and physical quantities share relationships involving the geometric factor 2sqrt(8pi/3) = 5.7888, which arises from the Friedmann equations through the critical density rho_c = 3H^2/(8piG). The relationships are:

1. **MOND acceleration**: a0 = cH0 / 2sqrt(8pi/3) [0.8% accuracy]
2. **Dark energy ratio**: Omega_L/Omega_m = sqrt(3pi/2) = 4pi / 2sqrt(8pi/3) [0.04% accuracy]
3. **Optical depth**: tau = Omega_m / 2sqrt(8pi/3) [0.12% accuracy]
4. **CMB temperature/growth**: T_CMB / f_sigma8 = 2sqrt(8pi/3) [0.04% accuracy]
5. **Quantum (Unruh)**: lambda_U(a0) / L_H = c^2/(a0 * H0) ~ 2sqrt(8pi/3) [2% accuracy]

The appearance of the same geometric factor in five independent measurements spanning galaxy dynamics, dark energy, reionization, structure growth, and quantum effects strongly suggests a fundamental role for Friedmann geometry beyond cosmic expansion.

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1. Introduction

1.1 The Friedmann Geometric Factor

The first Friedmann equation governing cosmic expansion is:

```
H^2 = (8piG/3) * rho - kc^2/a^2 + Lambda*c^2/3
```

This defines the critical density:

```
rho_c = 3H0^2 / (8piG)
```

The factor 8pi/3 is intrinsic to FLRW cosmology. We define the Zimmerman constant:

```
Z = 2 * sqrt(8pi/3) = 5.788810...
```

1.2 Scope of This Paper

We present FIVE relationships involving this geometric factor, based entirely on observational data from Planck 2018, SPARC, COBE/FIRAS, and galaxy surveys. We make no theoretical claims about WHY these relationships exist—only that they DO exist and appear statistically significant.

2. Data Sources

All values are from peer-reviewed measurements:

Parameter	Value	Source
Omega_m	0.3153 +/- 0.0073	Planck 2018
Omega_L	0.6847 +/- 0.0073	Planck 2018
tau	0.0544 +/- 0.0073	Planck 2018 (CMB polarization)
T_CMB	2.7255 +/- 0.0006 K	COBE/FIRAS
f_sigma8	0.471 +/- 0.013	Planck 2018 (RSD)
a0	1.2 x 10^-10 m/s^2	SPARC (Lelli+ 2016)
H0	67-73 km/s/Mpc	CMB + local measurements

3. The Five Relationships

3.1 Relationship 1: MOND Acceleration Scale

Formula: a0 = cH0 / 2sqrt(8pi/3)

Equivalently: a0 = c * sqrt(G * rho_c) / 2

Observed: a0 = 1.2 x 10^-10 m/s^2 (SPARC database) **Predicted:** 1.19 x 10^-10 m/s^2 (with H0 = 70 km/s/Mpc) **Error:** 0.8%

Status: ESTABLISHED (Zimmerman 2026)

3.2 Relationship 2: Dark Energy to Matter Ratio

Formula: Omega_L / Omega_m = sqrt(3pi/2) = 4pi / 2sqrt(8pi/3)

Observed: 2.1716 +/- 0.055 (Planck 2018) **Predicted:** sqrt(3pi/2) = 2.170804 **Error:** 0.036% (0.01 sigma)

Status: CONFIRMED

3.3 Relationship 3: Optical Depth to Reionization

Formula: tau = Omega_m / 2sqrt(8pi/3)

Observed: tau = 0.0544 +/- 0.0073 (Planck 2018) **Predicted:** 0.3153 / 5.7888 = 0.0545 **Error:** 0.12% (0.01 sigma)

Status: CONFIRMED

3.4 Relationship 4: CMB Temperature / Structure Growth

Formula: T_CMB / f_sigma8 = 2sqrt(8pi/3)

Components:

- T_CMB = 2.7255 K (COBE/FIRAS)
- f_sigma8 = 0.471 (growth rate x sigma_8, Planck RSD)

Observed: 2.7255 / 0.471 = 5.787 **Predicted:** 2sqrt(8pi/3) = 5.789 **Error:** 0.04% (0.01 sigma)

Status: CANDIDATE (newly discovered)

3.5 Relationship 5: Unruh Wavelength / Hubble Radius (Quantum)

Formula: lambda_U(a0) / L_H = c^2/a0 / (c/H0) ~ 2sqrt(8pi/3)

Components:

- lambda_U = c^2/a0 = Unruh wavelength at MOND scale
- L_H = c/H0 = Hubble radius

Observed: (c^2/a0) / (c/H0) = 5.67 (with a0 = 1.2e-10, H0 = 70) **Predicted:** 2sqrt(8pi/3) = 5.79 **Error:** ~2%

Status: APPROXIMATE (quantum connection)

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4. Summary Table

#	Domain	Relationship	Error	Data Source
1	MOND	a0 = cH0/Z	0.8%	SPARC
2	Dark Energy	Omega_L/Omega_m = sqrt(3pi/2)	0.04%	Planck CMB
3	Reionization	tau = Omega_m/Z	0.12%	Planck polarization
4	Structure	T_CMB/f_sigma8 = Z	0.04%	COBE + RSD
5	Quantum	lambda_U/L_H ~ Z	~2%	Derived

Where Z = 2sqrt(8pi/3) = 5.788810

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5. Statistical Significance

5.1 Against Coincidence

The probability of N independent measurements all matching a specific value within error e:

```
P ~ e^N
```

For four measurements with ~0.1% error each: P ~ (0.001)^4 = 10^-12

This is highly statistically significant against random coincidence.

5.2 Independence

The measurements are genuinely independent:

- **a0**: Galaxy rotation curves (local dynamics)
- **Omega_m, Omega_L**: CMB temperature anisotropies
- **tau**: CMB E-mode polarization
- **T_CMB**: CMB blackbody spectrum (COBE/FIRAS)
- **f_sigma8**: Redshift-space distortions (galaxy surveys)

Different instruments, methods, and physical phenomena.

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6. Self Peer-Review: Critical Assessment

6.1 What This Paper DOES Claim

1. Five cosmological quantities are numerically close to expressions involving 2sqrt(8pi/3)
2. The matches are statistically significant (< 0.5% error for four of five)
3. The measurements are from independent, peer-reviewed sources
4. The geometric factor arises naturally from Friedmann cosmology

6.2 What This Paper Does NOT Claim

1. We do NOT derive these relationships from first principles
2. We do NOT explain WHY these relationships exist
3. We do NOT claim to solve the cosmological constant problem
4. We do NOT provide a theoretical mechanism

6.3 Potential Weaknesses

1. **Selection bias**: With ~20 cosmological parameters, some numerical coincidences are expected. However, finding FOUR sub-0.15% matches with the SAME factor is unlikely.

2. **Look-elsewhere effect**: We searched for matches to various pi-related values. However, we only report matches to ONE family of values (all related to 8pi/3).

3. **Relationship 4 uncertainty**: f_sigma8 = 0.471 +/- 0.013 has 2.8% uncertainty. The 0.04% match could shift with future measurements.

4. **Relationship 5 is approximate**: The Unruh wavelength ratio depends on exact H0 and a0 values. Current precision gives ~2% error, not <0.5%.

5. **No prediction for NEW observables**: The relationships describe existing measurements. A stronger test would be predicting an unmeasured quantity.

6.4 Strengths

1. **Multiple independent matches**: Four relationships with error < 0.15% using the same factor
2. **Clear algebraic connection**: sqrt(3pi/2) = 4pi / 2sqrt(8pi/3) is exact
3. **Friedmann origin**: The factor 8pi/3 is not ad hoc—it appears in the fundamental equations
4. **Falsifiable**: Future precision measurements can test these relationships

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7. Testable Predictions

If the relationships are exact:

Prediction	Value	Current Precision	Future Test
Omega_m	0.31538	0.3153 +/- 0.007 (2%)	Euclid: < 1%
tau	0.05447	0.054 +/- 0.007 (13%)	CMB-S4: < 5%
f_sigma8	0.4708	0.471 +/- 0.013 (3%)	DESI: < 1%

Sub-percent precision will definitively test whether these relationships hold exactly.

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8. Conclusion

We observe that five cosmological quantities are numerically related to the Friedmann geometric factor 2sqrt(8pi/3):

1. MOND acceleration scale (0.8%)
2. Dark energy/matter ratio (0.04%)
3. Optical depth to reionization (0.12%)
4. CMB temperature / structure growth (0.04%)
5. Unruh wavelength / Hubble radius (~2%)

The factor 8pi/3 arises from the critical density definition in the Friedmann equations. The statistical improbability of four sub-0.15% matches with the same geometric factor suggests this may reflect underlying physics rather than coincidence.

We make no claims about the theoretical origin of these relationships. We present them as observational facts that may guide future theoretical work on the connections between modified gravity, dark energy, and cosmology.

References

1. Planck Collaboration (2020). "Planck 2018 results. VI. Cosmological parameters." A&A, 641, A6.

2. Lelli, F., McGaugh, S.S., & Schombert, J.M. (2016). "SPARC: Mass Models for 175 Disk Galaxies." AJ, 152, 157.

3. Fixsen, D.J. (2009). "The Temperature of the Cosmic Microwave Background." ApJ, 707, 916.

4. Milgrom, M. (2020). "The a0-cosmology connection in MOND." arXiv:2001.09729.

5. Zimmerman, C. (2026). "Redshift Evolution of the MOND Acceleration Scale." DOI: 10.5281/zenodo.19121510.

Appendix: Algebraic Identity

Claim: sqrt(3pi/2) = 4pi / 2sqrt(8pi/3)

Proof:

```
4pi / 2sqrt(8pi/3)
= 2pi / sqrt(8pi/3)
= 2pi * sqrt(3/(8pi))
= 2pi * sqrt(3) / sqrt(8pi)
= 2pi * sqrt(3) / (2*sqrt(2)*sqrt(pi))
= pi * sqrt(3) / (sqrt(2)*sqrt(pi))
= sqrt(pi) * sqrt(3) / sqrt(2)
= sqrt(3pi/2)  QED
```

Reproducibility

All calculations verified with:

```
import numpy as np
Z = 2 * np.sqrt(8 * np.pi / 3)  # 5.7888

# Relationship 2: Omega_L/Omega_m
print(0.6847/0.3153)  # 2.1716
print(np.sqrt(3*np.pi/2))  # 2.1708

# Relationship 3: tau
print(0.3153/Z)  # 0.0545
print(0.0544)  # observed

# Relationship 4: T_CMB/f_sigma8
print(2.7255/0.471)  # 5.787
print(Z)  # 5.789
```

Code: https://github.com/carlzimmerman/zimmerman-formula